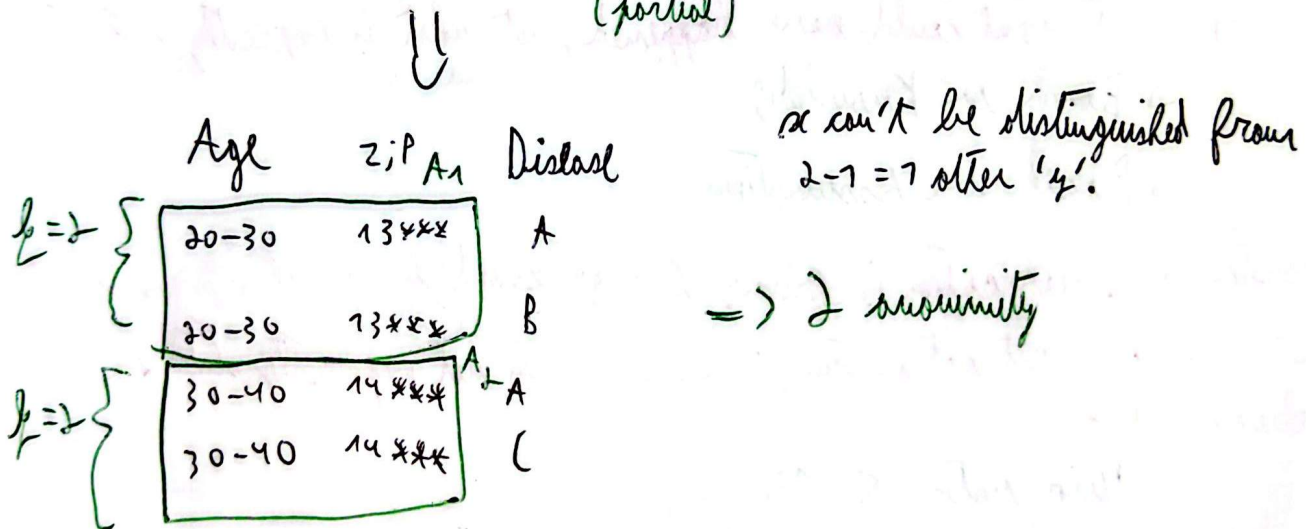
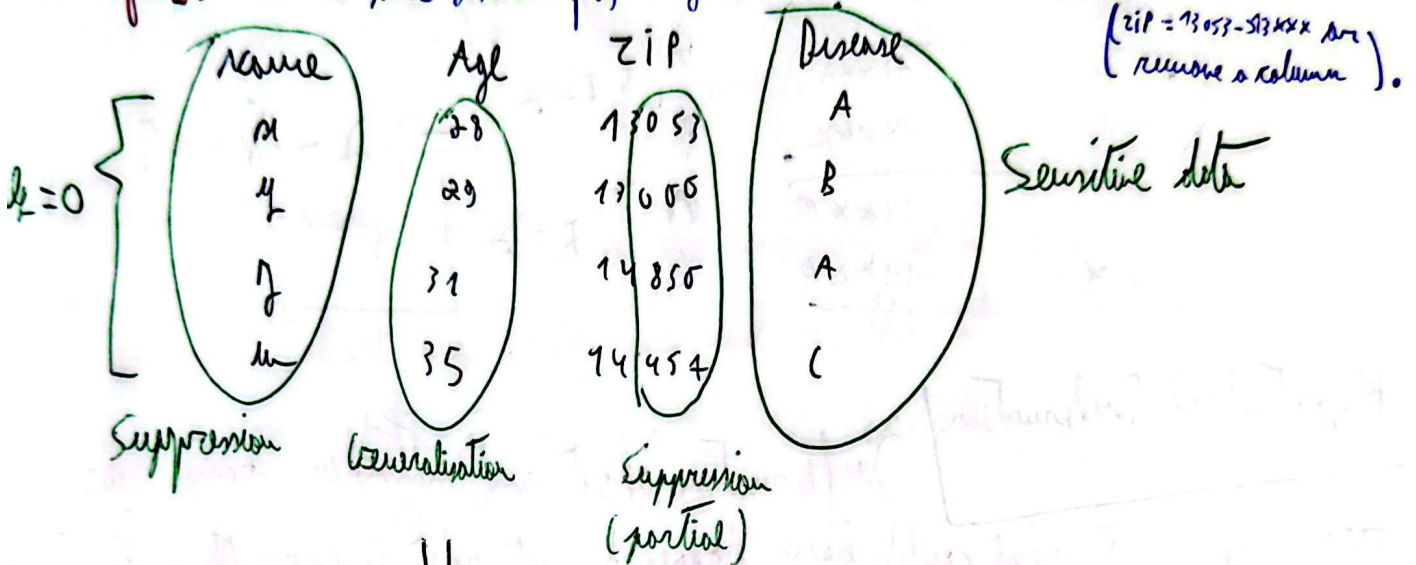


Privacy:

0.0

(HAPS:

K-Anonymity } A dataset is k -anonymous if each individual row is indistinguishable from at least $k-1$ others with respect to the quasi-identifiers. There are 2 Techniques: generalisation (age = 22-35 age < 24) or suppression (zip = 13053-53xxx are remove a column).



k -anonymity protects against identity disclosure, but it does not always protect sensitive attributes. If we know m has (disease), he is not protected at all.

I-Diversity | A dataset satisfies I-diversity if, for every every equivalence class, there are at least I "well-represented" distinct values of the sensitive attribute. 5.1

- ▷ Equivalence Class = Records with the same quasi-identifiers.
- ▷ Sensitive Attribute = Attribute to well distribute.

Age	ZIP	Disease
20-30	13***A	A
20-30	13***X	B
30-40	14***A	A
30-40	14***B	C

$I=2 \Rightarrow 2\text{-diversity}$
 $I=2$

Hypothetical Information | Information about an attacker can infer by reasoning about what could have happened, not what is explicitly in the dataset.

- ▷ Based on Knowledge
- ▷ Based on Assumptions

Hypothetical information is privacy leakage caused by logical inference about what must or must not be true, even when it is not explicitly stated in the released data.

Voic notes 5.19

"If I suppose something on someone, then I can approximate someone else."

Re-identification levels

- 1: $u = u_i$
- 2: $u \in \{u_i, u_j, \dots\}$
- ...

$$\tilde{H}(U; V) = H[U] + \sum_{A \in V} P_A[U] \cdot \sum_{B \in DB} \tilde{P}_B[A|U] \cdot \log_2 \tilde{P}_B[U|A]$$

$$P(X=M) = P(M) \wedge H[U] = - \sum_u P[U] \cdot \log_2 P[U]$$

Perceived information

8.2

Is the information an attacker believes or learns after seeing the released data, given their prior knowledge.

- what attacker infers.
- How beliefs change. → About an individual.
- what is understood.

$$P_I^1(U; O) = H[U] + \sum_{u \in U} P[u] \cdot \sum_{o \in DB_U} \frac{1}{|DB_U|} \cdot \log_2 \frac{P[u|o]}{P[u]}$$

Differential Privacy

Ensures the output of a computation does not reveal much about any single individual in the dataset.

- If your data is in or out, the results of an analysis should look almost the same.
- An attacker cannot confidently tell if a specific person is in the dataset or not.
- Unlike k -anonymity or I -diversity, it does not rely on syntactic transformations but it gives a mathematical guarantee.

MPC & FHE

MPC $\equiv$ Multi-Party Computation: Allows multiple parties to jointly compute a function over their inputs without revealing their private inputs to each other.

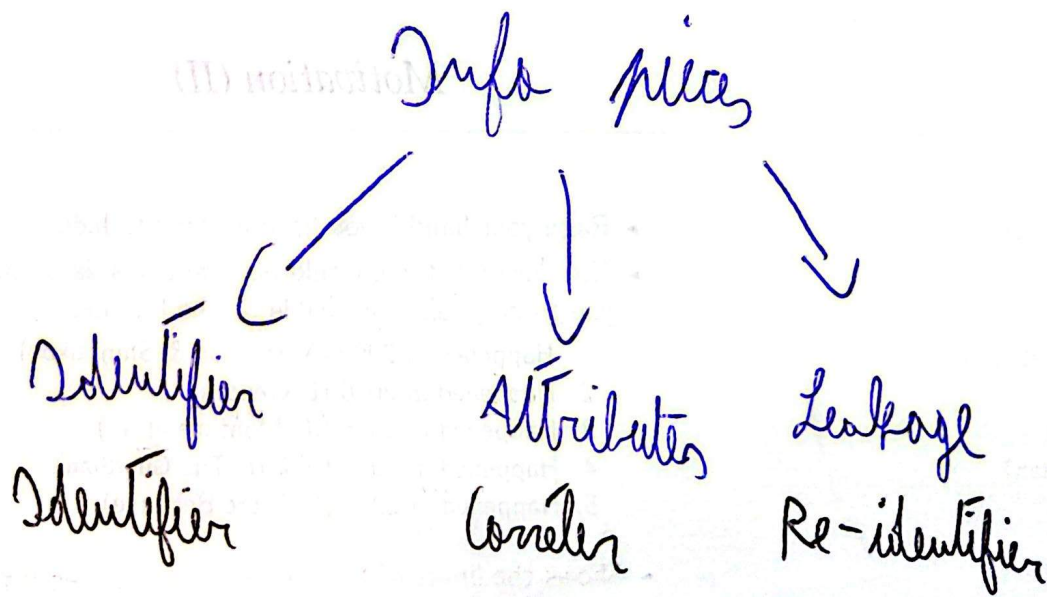
FHE $\equiv$ Fully Homomorphic Encryption: Allows computation on encrypted data and the decryption is the result of all executed operations.

FHE-SE $\equiv$ with Secure Enclaves.

Basics

S9: Fair information practices:

- > Existence of personal data storage cannot be secret.
- > Individuals can access their data.
- > Data minimization principle.
- > Need of consent (opt in / out data collection).
- > Automated (pseudo) decisions can be contested.
- > Organizations are responsible of data leaks.



S.15

2 rows because each group needs at least 3 sensitive data.

Ideal:

- ▷ Ideal $\mathbb{F}$ -diversity means uniform and strong diversity: every equivalence class has to all possible disease values.
- ▷ with discrete sensitive values, assume 4 diseases total.
- ▷ Each group currently has 2, so needs 2 more per group.
- ▷ With 4 equivalence classes after balancing, this leads to 8 additional rows.

(S.15)

K-Anonymity:

- ▷ Removes sensitive attributes.
- ▷ Suppress or generalize some info.
- ▷ Count the number of min similarity.
- ▷ Include someone in this has common correlation attribute values.
- ▷ Not enough alone needs $\mathbb{F}$ -diversity.

$\mathbb{I}$ - Diversity:

- ▷ well distributed sensitive data equally between the k -anonymous groups.
- ▷ At least $\mathbb{I}$ distinct values in each $A(s)$.

Pseudonyms

- Example of "pseudonymized" deterministic database

pseudo.	job	sex	age	category
U1	engineer	m	35	happily
U2	engineer	m	35	happily
U3	lawyer	m	35	hi
U4	writer	f	30	hi
U5	writer	f	30	hi
U6	dancer	f	30	hi
U7	dancer	f	30	hi

- How strong are pseudonyms if we learn/leak that:
 - Cathy is a 30 yo, female writer in the DB?

U4 or U5

Pseudonyms

- Example of "pseudonymized" deterministic database

pseudo.	job	sex	age	category
U1	engineer	m	35	happily
U2	engineer	m	35	happily
U3	lawyer	m	35	hi
U4	writer	f	30	hi
U5	writer	f	30	hi
U6	dancer	f	30	hi
U7	dancer	f	30	hi

- How strong are pseudonyms if we learn/leak that:
 - Emily is a 30 yo, female dancer in the DB?

U6 or U7

Pseudonyms

- Example of "pseudonymized" deterministic database

pseudo.	job	sex	age	category
U1	engineer	m	35	happily
U2	engineer	m	35	happily
U3	lawyer	m	35	hi
U4	writer	f	30	hi
U5	writer	f	30	hi
U6	dancer	f	30	hi
U7	dancer	f	30	hi

- Would you prefer being Cathy or Emily? {...}

Cathy leaked less info so it was better.

Pseudonyms

- Example of "pseudonymized" deterministic database

pseudo.	job	sex	age	category
U1	engineer	m	35	happily
U2	engineer	m	35	happily
U3	lawyer	m	35	hi
U4	writer	f	30	hi
U5	writer	f	30	hi
U6	dancer	f	30	hi
U7	dancer	f	30	hi

- Would you prefer being Cathy or Emily? {...}
- How would you improve the situation? {...}

k-anonymity (I)

- Aims at preventing linkage on pseudonyms
- e.g., thanks to information leakage (or simply non-anonymized/non-sensitive databases)

- Let o be a row of the database, u be a user
- Define anonymity set: $A(o) = \{u | DB(u, :) = o\}$
- k -anonymity is provided for

$$k = \min_{o \in DB} |A(o)|$$

Set of users such that they have the exactly same sensitive attribute

- Typically achieved with 3 technical tools, namely

- Aggregation of attributes
- Grouping users
- Noise addition (no data)

k-anonymity (II)

- Example of "k-anonymized" deterministic database

pseudo.	job	sex	age	category
U1	professional	m	35-40	happily
U2	professional	m	35-40	happily
U3	professional	m	35-40	hi
U4	artist	f	30-35	hi
U5	artist	f	30-35	hi
U6	artist	f	30-35	hi
U7	artist	f	30-35	hi

- Which tool does it use?

✖ All min(n -occurrences)
should be = current I

$$I_0 = (2, 1, 0)$$

$$I_1 = (2, 3, 0)$$

$$I = 2 \Rightarrow I = (2, 2, 2)$$

$$I_0 = (0, 1, 2)$$

$$I_1 = (3, 0, 2)$$

→ 8 rows

*

What is the value of I ? 2

2 rows

How many records needed to reach $I = 3$?

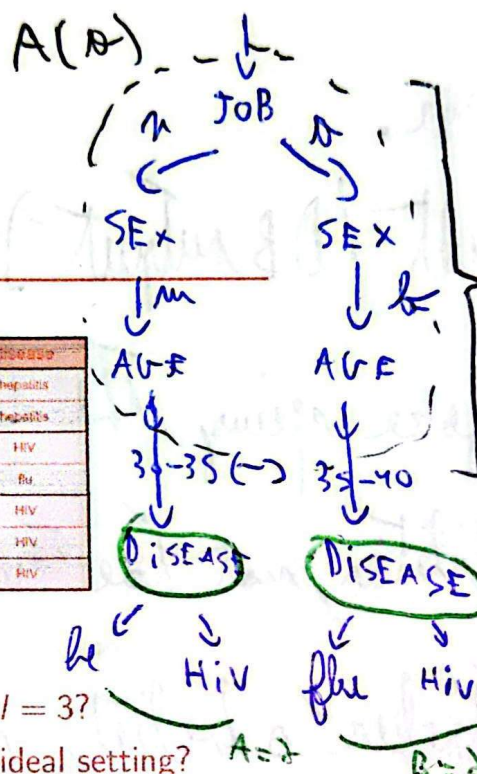
How many records needed to reach ideal setting?

What are the main limitations of this setting? {...}

All values are discrete

I -diversity (II)

pseudo.	job	sex	age	disease
u1	professional	m	35-40	hepatitis
u2	professional	m	35-40	hepatitis
u3	professional	m	35-40	HIV
u4	artist	f	30-35	flu
u5	artist	f	30-35	HIV
u6	artist	f	30-35	HIV
u7	artist	f	30-35	HIV



Interlude: t -closeness

Firstly
all non-sensitive
arguments

- Formalizes a relaxation of the "ideal setting"
- The distribution of sensitive attributes in an anonymity set is "close" to the distribution of the full table
 - Close is captured by a statistical distance
- We will discuss it together with the second model

$$I = \min(A, B) = 2$$

Records are grouped by quasi-identifiers (job, sex, age).

Each group must contain at least I different sensitive values.

$$(prof, m, 35-40) \Rightarrow \{HIV, hepatitis\}$$

$$(artist, f, 30-35) \Rightarrow \{HIV, flu\}$$

Probabilistic databases

Probabilistic databases

pseudo.	comedy	drama

pseudo.	comedy	drama

5.19

- V : Random variable = which user.
- O : Observation (query result / DB output).
- $H[V]$: Entropy of the user before seeing the DB.
- $\tilde{Pr}$: Attacker's estimated probabilities, not the true ones.

This measures how much the attacker's beliefs shift and how if those beliefs are wrong. $\Rightarrow$ So $H(V; O)$ is hypo-hypothetical.

~~Pr[User] = Pr[Comedy] = Pr[Drama]~~

Hypothetical information is imagined or what-if data used to explore possible outcomes, with the point of predicting, reasoning, or decision-making without needing real-world evidence.

EX: I have no access to the DB but I assume there are 50% of men.

$H I = 0 \Rightarrow$ No info / clues.

$H I > 0 \Rightarrow$ Potentially we can infer V from O .

Assumed: (not verified as no access to DB)

	comedy	drama
μ_1	55%	35%
μ_2	50%	50%

$$H I(V; O) = 1 + 0,5 \cdot [0,55 \cdot \log_2(0,505) + 0,35 \cdot \log_2(0,412)] + 0,5 \cdot [0,5 \cdot \log_2(0,455) + 0,5 \cdot \log_2(0,588)] = 0,014$$

$$H[V] = -Pr[\mu_1] \cdot \log_2 Pr[\mu_1] - Pr[\mu_2] \cdot \log_2 Pr[\mu_2] = 1$$

$$Pr[\mu_1] = Pr[\mu_2] = 0,5$$

$$Pr^{true}[comedy | \mu_1] = 0,55$$

$$Pr^{true}[drama | \mu_1] = 0,35$$

$$Pr^{true}[comedy | \mu_2] = 0,5$$

$$Pr^{true}[drama | \mu_2] = 0,5$$

$$Pr^{est}[\mu_1 | comedy] = \frac{0,55}{(0,55+0,5)} = 0,505$$

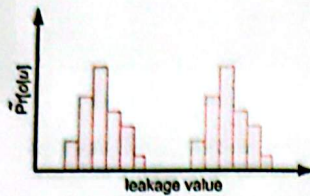
$$Pr^{est}[\mu_1 | drama] = \frac{0,35}{(0,35+0,5)} = 0,412$$

$$Pr^{est}[\mu_2 | comedy] = 1 - Pr^{est}[\mu_1 | comedy] = 0,455$$

$$Pr^{est}[\mu_2 | drama] = 1 - Pr^{est}[\mu_1 | drama] = 0,588$$

Intuition

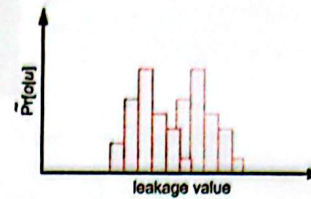
- Consider a 2-user case study with $\tilde{P}_r[o|u]$



- It leads to a Hypothetical Information $\tilde{H}(U; O) = 1$

Intuition

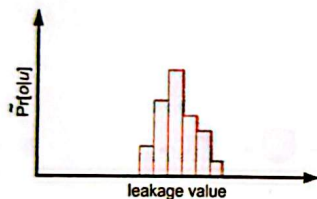
- Consider a 2-user case study with $\tilde{P}_r[o|u]$



- It leads to a Hypothetical Information $\tilde{H}(U; O) \in]0, 1[$

Intuition

- Consider a 2-user case study with $\tilde{P}_r[o|u]$



- It leads to a Hypothetical Information $\tilde{H}(U; O) = 0$

Cautionary notes

- Analysis per user since they can have very different distributions (more or less discriminating)
- Assumptions most likely do not hold for actual databases (actual user's data does not come from a regular distribution and is not enough noisy)

Cautionary notes

- Analysis per user since they can have very different distributions (more or less discriminating)
- Assumptions most likely do not hold for actual databases (actual user's data does not come from a regular distribution and is not enough noisy)
- Yet, general intuition is correct: users' privacy decreases exponentially with the number of leakages
 - Privacy vanishes fast in open data publishing
 - & things get worse if multiple DBs can be combined

From Orwell to Kafka



Kafka's metaphor: surreal nightmare of a person unexpectedly informed that he is under arrest but given no reason why

- Looks far sketched?

5.20

$$\begin{aligned} * \hat{P}^{\sim}(v; 0) &= H(v) + \sum_u Pr(v) \left\{ \frac{1}{|DB_K|} \cdot \log_2(\hat{P}^{\sim}[u|v] = \hat{P}^{\sim}[u]) \right\} \\ &= H(v) + \sum_u Pr(v) \cdot |DB_K| \cdot \frac{1}{|DB_K|} \cdot \log_2 \hat{P}^{\sim}[u] = 0 \end{aligned}$$

$a \in \{\text{comedy, drama}\}$

$$= -H(v)$$

The attacker has no access to DB_K but only to DB_P so he does not see any differences between u_1 and u_2 .

** For more samples and better distribution

$$\hat{P}^{\sim}[u_1 | \text{comedy}] = \frac{110}{110+105} = 0,512$$

$$\hat{P}^{\sim}[u_1 | \text{drama}] = \frac{90}{90+95} = 0,485$$

$$\hat{P}^{\sim}[u_2 | \text{comedy}] = 1 - \hat{P}^{\sim}[u_1 | \text{comedy}] = 0,488$$

$$\hat{P}^{\sim}[u_2 | \text{drama}] = 1 - \hat{P}^{\sim}[u_1 | \text{drama}] = 0,514$$

$$Pr[u_1] = \frac{(50+40)}{(50+40+55+45)} = 0,5$$

$$Pr[u_2] = 1 - Pr[u_1] = 0,5$$

$$\begin{aligned} & \frac{209}{209+251} = 0,544 \\ & = 0,402 \end{aligned}$$

$$= 1 - 0,544 = 0,456$$

$$= 1 - 0,402 = 0,598$$

$$= 0,5$$

$$= 0,5 \Rightarrow \hat{P}^{\sim}(v, 0) \approx 0,009$$

$$\Rightarrow \hat{P}^{\sim}(v, 0) = \left(-Pr(u_1) \cdot \log_2(Pr(u_1)) - Pr(u_2) \cdot \log_2(Pr(u_2)) \right)$$

$$+ Pr(u_1) \cdot \frac{1}{|DB_K|} \cdot \left[\log_2 \hat{P}^{\sim}[u_1 | \text{comedy}] + \log_2 \hat{P}^{\sim}[u_1 | \text{drama}] \right]$$

$$+ Pr(u_2) \cdot \frac{1}{|DB_K|} \cdot \left[\log_2 \hat{P}^{\sim}[u_2 | \text{comedy}] + \log_2 \hat{P}^{\sim}[u_2 | \text{drama}] \right]$$

$$= (-0,5 \cdot \log_2(0,5) - 0,5 \cdot \log_2(0,5)) + 0,5 \cdot \frac{1}{2} \cdot [\log_2(0,512) + \log_2(0,485)]$$

$$+ 0,5 \cdot \frac{1}{2} \cdot [\log_2(0,488) + \log_2(0,514)] \approx 0,0005 \approx 0$$

The attacker doesn't gain very much information.

Perceived Information

$\hat{P}I(U; O) \equiv$ what is perceived by the attacker so conclusions from DB_p.
 $\hat{P}I(U; O) \approx DB_p$

- Split the DB in two parts: DB_p and DB_t
 - Where p stands for profiling and t for testing
 - Build a model $\hat{P}r[o|u] \leftarrow DB_p$ for every user
- $$\hat{P}I(U; O) = H[U] + \sum_{u \in U} \Pr[u] \cdot \sum_{o \in DB_t} \frac{1}{|DB_t|} \cdot \log_2 \underbrace{\hat{P}r[u|o]}_{\text{model}}$$
- But true distr. unknown $\Rightarrow$ can only be sampled
 - Intuitively: amount of information that can be extracted from obs. biased by model errors

pseudo.	DB _p		DB _t	
	comedy	drama	comedy	drama
U1	88 (prob 88%)	45 (prob 45%)	888 (prob 88%)	488 (prob 48%)
U2	45 (prob 45%)	45 (prob 45%)	888 (prob 88%)	488 (prob 48%)

- With $|DB_p| = 100$, $\hat{P}I(U; O) = 0$

Impossible de distinguer
 $\hat{P}r(u|s) = \hat{P}r(u) *$

Examples

pseudo.	DB _p		DB _t	
	comedy	drama	comedy	drama
U1	118 (prob 67%)	88 (prob 47%)	888 (prob 87%)	488 (prob 47%)
U2	155 (prob 57%)	88 (prob 47%)	888 (prob 87%)	888 (prob 47%)

- With $|DB_p| = 100$, $\hat{P}I(U; O) = 0$
- What is the value of $\hat{P}I(U; O)$ with $|DB_p| = 200$ ~~***~~

Examples

pseudo.	DB _p		DB _t	
	comedy	drama	comedy	drama
U1	298 (prob 67%)	381 (prob 47%)	888 (prob 87%)	488 (prob 47%)
U2	251 (prob 57%)	348 (prob 47%)	888 (prob 87%)	888 (prob 47%)

- With $|DB_p| = 100$, $\hat{P}I(U; O) = 0$
- What is the value of $\hat{P}I(U; O)$ with $|DB_p| = 500$ ~~***~~

Examples

pseudo.	DB _p		DB _t	
	comedy	drama	comedy	drama
U1	388 (prob 67%)	381 (prob 47%)	888 (prob 87%)	488 (prob 47%)
U2	351 (prob 57%)	348 (prob 47%)	888 (prob 87%)	888 (prob 47%)

- With $|DB_p| = 100$, $\hat{P}I(U; O) = 0$
- What is the value of $\hat{P}I(U; O)$ with $|DB_p| = 500$
- As the size of DB_p increases, the model better reflects / predicts the true distribution $\Rightarrow$ re-identification attacks with external leakages become more effective

Summary (I)

- Re-identification with external leakages also possible
- But the biggest concern in this case is inference

Towards Model 2

- ▶ Model 1 is too liberal w.r.t. privacy
- ▶ How can we reduce the Adv.'s power?

- ▶ Only access to output of queries

- ▶ Ideal setting re-stated:

but outputs should not leak anything about an individual (that could not be learnt without accessing the DB).

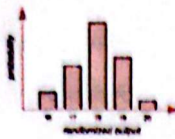
- ▶ Is it possible at all? {...}

↳ Not really so we should add noise.

Example

Tool: noise addition

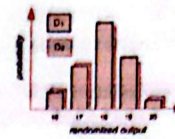
pseudo	smoker	cancer
01	Y	Y
02	Y	N
03	N	N
04	N	N
05	Y	Y
06	N	N
07	N	Y
...	...	...



- More realistic promise: if you leave the database, the (randomized) outcome will not change much
- So why leaving the experiment in this case?

Differential Privacy

pseudo	smoker	cancer
01	Y	Y
02	Y	N
03	N	N
04	N	N
05	Y	Y
06	N	N
07	N	Y
...	...	...

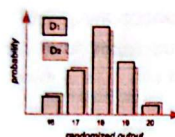


- ϵ -differential privacy for mechanism M if for any two neighboring DBs D_1, D_2 and any $c \in \text{range}(M)$

$$\Pr[M(D_1) = c] \leq e^\epsilon \Pr[M(D_2) = c]$$

Differential Privacy

pseudo	smoker	cancer
01	Y	Y
02	Y	N
03	N	N
04	N	N
05	Y	Y
06	N	N
07	N	Y
...	...	...



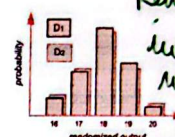
- ϵ -differential privacy for mechanism M if for any two neighboring DBs D_1, D_2 and any $c \in \text{range}(M)$

$$\Pr[M(D_1) = c] \leq (1 + \epsilon) \Pr[M(D_2) = c] + \epsilon$$

(Intuitively: probabilities are multiplicatively close)

Differential Privacy

pseudo	smoker	cancer
01	Y	Y
02	Y	N
03	N	N
04	N	N
05	Y	Y
06	N	N
07	N	Y
...	...	...



Re-sampling and individual does not change the output very much => Good!!

- (ϵ, δ) -differential privacy for mechanism M if for any two neighboring DBs D_1, D_2 and any $c \in \text{range}(M)$

$$\Pr[M(D_1) = c] \leq (1 + \epsilon) \Pr[M(D_2) = c] + \delta$$

(Intuitively: to deal with the outputs with $\Pr = 0$)

Noise quantification

- $\propto$ the sensitivity of the function f to evaluate

$$\Delta f = \max_{D_1, D_2} |f(D_1) - f(D_2)|$$

- i.e., how much one person can affect the output
- e.g., what is the sensitivity for counting queries that count number of rows satisfying a predicate?

DB P_1 has one person less than D_2 .

Noise quantification

- $\propto$ the sensitivity of the function f to evaluate

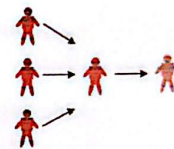
$$\Delta f = \max_{D_1, D_2} |f(D_1) - f(D_2)|$$

- i.e., how much one person can affect the output
- e.g., what is the sensitivity for counting queries that count number of rows satisfying a predicate?
- Theorem: query f hidden by Laplacian noise with parameter $b = \frac{\Delta f}{\epsilon}$ achieves ϵ -differential privacy

- What is the goal of biometric identification? {...}
- How many input/output parties does it involve?
- How to obtain the eigenface repr. of an image? {...}
- Why can profiling be excluded from the MPC? {...}
- Is the projection (eigenvectors) sensitive? {...}

- What is the goal of biometric identification? {...}
- How many input/output parties does it involve?
- How to obtain the eigenface repr. of an image? {...}
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- Is the projection (eigenvectors) sensitive? {...}
- Why is eigenf. repr. a good candidate for MPC? {...}

- What is the goal of biometric identification? {...}
- How many input/output parties does it involve? **2**
- How to obtain the eigenface repr. of an image? {...}
- Why can profiling be excluded from the MPC? {...}
- Is the projection (eigenvectors) sensitive? {...} **YES**
- Why is eigenf. repr. a good candidate for MPC? {...} **Exclusion Distance**
- How to protect against a dishonest out. party? **DP**



- One processing party (!), one or many input parties, usually one output party (but could be more)
- To protect against malicious computing party, need to add a verifiable computation protocol on top
- (+ DP needed if the output party is dishonest)

